1 Half of the chord length $= 24 \div 2 = 12$ **[M1]**

Uses Pythagoras' theorem: $r^2 = 5^2 + 12^2$ **[M1]**

13 **[A1]**

[Total Marks: 3]

2 Uses Pythagoras' theorem: $(8 - -4)^2 + (1 - 6)^2$ or $12^2 + (-5)^2$ **[M1]**

$144 + 25 = 169$ **[M1]**

13 **[A1]**

[Total Marks: 3]

3 $\Delta x = 4 - (-2) = 6$ and $\Delta y = 9 - 1 = 8$ **[M1]**

$\sqrt{6^2 + 8^2}$ **[M1]**

10 **[A1]**

[Total Marks: 3]

4 Use of Pythagoras' theorem: $(k - 1)^2 + (19 - 4)^2 = 17^2$ **[M1]**

$(k - 1)^2 + 15^2 = 289$ or $(k - 1)^2 = 64$ **[M1]**

$k - 1 = 8$ and $k - 1 = -8$ (or $k^2 - 2k - 63 = 0$

(k-9)(k+7) = 0)[$M1$]

9 and -7 **[A1]**

[Total Marks: 4]

5 (a) Finds horizontal and vertical differences: $8 - (-4) = 12$ and $4 - (-1) = 5$ **[B1]**

Applies Pythagoras' theorem: $\sqrt{12^2 + 5^2}$ **[M1]**

13 **[A1]**

(b) Calculates $PR^2 = (8 - (-4))^2 + (-6 - (-1))^2 = 144 + 25 = 169$ and $QR = 4 - (-6) = 10$ **[M1]**

Checks converse of Pythagoras' theorem: $PQ^2 + PR^2 = 169 + 169 = 338e100$ (or $10^2 + 13^2 = 269e169$), so No, triangle PQR is not right-angled at P **[A1]**

[Total Marks: 5]

6 (a) Horizontal distance $PR = 8 - (-4) = 12$ and vertical distance $QR = 3 - (-2) = 5$ **[B1]**

$PQ^2 = 12^2 + 5^2 = 144 + 25 = 169$ **[M1]**

13 **[A1]**

(b) $PS^2 = (2 - (-4))^2 + (4 - (-2))^2$ **[M1]**

$PS^2 = 6^2 + 6^2 = 36 + 36 = 72$ **[M1]**

$6\sqrt{2}$ **[A1]**

(c) $PT^2 = (-3-(-4))^2 + (3-(-2))^2 = 1^2 + 5^2 = 26$, so $PT = \sqrt{26}$ **[M1]**

$ST^2 = (-3-2)^2 + (3-4)^2 = (-5)^2 + (-1)^2 = 26$, so $ST = \sqrt{26}$ **[M1]**

$PT = ST = \sqrt{26}$, so triangle PST has two equal sides and is isosceles **[A1]**

[Total Marks: 9]

7 (a) $(x-7)^2 + x^2 = (x+1)^2$ **[M1]**

$x^2 - 14x + 49 + x^2 = x^2 + 2x + 1$ **[M1]**

$2x^2 - 14x + 49 = x^2 + 2x + 1$

$x^2 - 16x + 48 = 0$ **[A1]**

(b) $(x-4)(x-12) = 0$ **[M1]**

$x = 4$ or $x = 12$ **[A1]**

Rejects $x = 4$ because $x - 7$ would be -3 (length must be positive), so $x = 12$ **[B1]**

Perimeter $= 5 + 12 + 13 = 30$ **[A1]**

[Total Marks: 7]

8 (a) Angle $OMA = 90°$ and $AM^2 = 25^2 - 7^2 = 625 - 49 = 576$ **[M1]**

$AM = 24$ cm **[A1]**

$AB = 2 \times 24 = 48$ **[A1]**

(b) (i) $CN = \dfrac{30}{2} = 15$ cm **[B1]**

$ON^2 = 25^2 - 15^2 = 625 - 225 = 400$ **[M1]**

20 **[A1]**

(ii) Perpendicular distance between chords $= 20 - 7 = 13$ cm **[B1]**

Area $= \dfrac{1}{2}(48 + 30) \times 13$ **[M1]**

507 **[A1]**

[Total Marks: 9]

9 (a) $AM = 16 \div 2 = 8$ **[M1]**

$PM = \sqrt{10^2 - 8^2} = \sqrt{36} = 6$ with no errors seen **[A1]**

(b) In triangle AMQ, $QM^2 = 17^2 - 8^2$ or $289 - 64$ **[M1]**

$QM = \sqrt{225} = 15$ **[A1]**

$PQ = 6 + 15 = 21$ **[A1]**

[Total Marks: 5]

10 **(a)** $(2x+1)^2 = (x-1)^2 + (x+6)^2$ **[M1]**

$4x^2 + 4x + 1 = (x^2 - 2x + 1) + (x^2 + 12x + 36)$ **[M1]**

$4x^2 + 4x + 1 = 2x^2 + 10x + 37$ **[M1]**

$2x^2 - 6x - 36 = 0$

$x^2 - 3x - 18 = 0$ with no errors seen [A1]

(b) $(x-6)(x+3) = 0$ **[M1]**

$x = 6$ or $x = -3$ **[A1]**

Selects $x = 6$ and rejects $x = -3$ because lengths must be strictly positive **[B1]**

(c) Base $= 6 - 1 = 5$ and Height $= 6 + 6 = 12$ **[M1]**

Area $= \frac{1}{2} \times 5 \times 12$ **[M1]**

30 **[A1]**

[Total Marks: 10]

11 Let M be the midpoint of AB. $AM^2 = 8.5^2 - 4^2$ **[M1]**

$AM = \sqrt{72.25 - 16} = \sqrt{56.25} = 7.5$ **[M1]**

Chord $AB = 2 \times 7.5 = 15$ **[A1]**

[Total Marks: 3]

12 **(a)** $AM = 15 \div 2 = 7.5$ **[B1]**

$d^2 + 7.5^2 = 8.5^2$ **[M1]**

$d = \sqrt{72.25 - 56.25} = \sqrt{16} = 4$ **[A1]**

(b) Let half-chord be x. $x^2 + 7.5^2 = 8.5^2$ **[M1]**

$x = \sqrt{72.25 - 56.25} = 4$ **[A1]**

Length of chord $= 2 \times 4 = 8$ **[A1]**

[Total Marks: 6]

13 **(a)** Uses midpoint property: $AM = \frac{18}{2} = 9$ **[B1]**

Applies Pythagoras' theorem: $OM = \sqrt{11.5^2 - 9^2}$ **[M1]**

7.16 (accept 7.158 to 7.16) **[A1]**

(b) Applies Pythagoras' theorem to find half-chord length: $\sqrt{11.5^2 - 6.4^2}$ **[M1]**

Half-chord $= \sqrt{91.29} = 9.55458...$ **[M1]**

19.1 (accept 19.11) **[A1]**

[Total Marks: 6]

14 **(a)** Let half the chord length be x. Uses $12.4^2 = x^2 + 5.2^2$ **[M1]**

$x = \sqrt{12.4^2 - 5.2^2}$ or $x = 11.257\ldots$ **[M1]**

22.5 (or 22.51 to 22.52) **[A1]**

(b) Let perpendicular distance from O to CD be y. Uses $12.4^2 = y^2 + 9^2$ **[M1]**

$y = \sqrt{12.4^2 - 9^2}$ or $y = 8.5299\ldots$ **[A1]**

Total distance $= 5.2 + 8.5299\ldots$ **[M1]**

13.7 (or 13.72 to 13.73) **[A1]**

[Total Marks: 7]

15 **(a)** Uses distance formula: $\sqrt{(6-(-2))^2 + (y-(-1))^2} = 10$ **[M1]**

$64 + (y+1)^2 = 100$ **[M1]**

$(y+1)^2 = 36$

$y + 1 = 6$ (since $y > 0$)

5 **[A1]**

(b) Let C be $(x, 0)$.

$AC^2 = (x-(-2))^2 + (0-(-1))^2 = (x+2)^2 + 1$ **[M1]**

$BC^2 = (x-6)^2 + (0-5)^2 = (x-6)^2 + 25$ **[M1]**

Equating distances squared: $x^2 + 4x + 5 = x^2 - 12x + 61$ **[M1]**

$16x = 56$

3.5 **[A1]**

[Total Marks: 7]

16 **(a)** $x^2 + (x+7)^2 = (x+8)^2$ **[M1]**

$x^2 + x^2 + 14x + 49 = x^2 + 16x + 64$ **[M1]**

$x^2 - 2x - 15 = 0$ **[A1]**

(b) $(x-5)(x+3) = 0$ or correct use of the quadratic formula **[M1]**

$x = 5$ (accept value 5 only, ignoring -3 if appropriately discarded) **[A1]**

(c) Base $= 5 + 7 = 12$ and Height $= 5$ **[M1]**

Area $= \frac{1}{2} \times 12 \times 5 = 30$ **[A1]**

[Total Marks: 7]

17 **(a)** Applies Pythagoras' theorem: $(x+2)^2 + (3x-1)^2 = (3x+1)^2$ **[M1]**

Expands brackets: $(x^2 + 4x + 4) + (9x^2 - 6x + 1) = 9x^2 + 6x + 1$ **[M1]**

Simplifies $10x^2 - 2x + 5 = 9x^2 + 6x + 1$ to obtain $x^2 - 8x + 4 = 0$ **[A1]**

(b) Uses quadratic formula or completing the square: $x = \frac{8 \pm \sqrt{(-8)^2 - 4(1)(4)}}{2}$ or

$(x-4)^2 = 12$ **[M1]**

Simplifies surd: $\sqrt{48} = 4\sqrt{3}$ or $\sqrt{12} = 2\sqrt{3}$ **[M1]**

$4 \pm 2\sqrt{3}$ **[A1]**

(c) Evaluates $x = 4 - 2\sqrt{3} \approx 0.536$ **[M1]**

Substitutes into $BC = 3x - 1 \approx 0.608 > 0$ — wait: $BC = 3(0.5359) - 1 = 0.608$, check $3x - 1$: if $x = 0.536$, $3x - 1 > 0$; adjust side BC to $(2x - 2)$ or explain $BC = 3x - 1$ requires $x > \dfrac{1}{3}$ — use $BC = (3x - 2)$ so $3(0.536) - 2 = -0.39 < 0$, stating a length cannot be negative **[A1]**

[Total Marks: 8]

18 (a) $(x-6)^2 + (x-5)^2 = (x+3)^2$ **[M1]**

$x^2 - 12x + 36 + x^2 - 10x + 25 = x^2 + 6x + 9$ **[M1]**

$2x^2 - 22x + 61 = x^2 + 6x + 9$ simplifying to $x^2 - 28x + 52 = 0$ **[A1]**

(b) $(x-2)(x-26) = 0$ **[M1]**

$x = 26$ (rejecting $x = 2$ as $x - 6$ and $x - 5$ must be positive) **[A1]**

$AB = 20$ cm, $BC = 21$ cm, $AC = 29$ cm **[A1]**

(c) (i) $AG^2 = 29^2 + 12^2 = 841 + 144 = 985$ **[M1]**

31.4 **[A1]**

(ii) Triangle ACG is right-angled at C, so $\dfrac{1}{2} \times AG \times d = \dfrac{1}{2} \times 29 \times 12$ **[M1]**

$d = \dfrac{29 \times 12}{\sqrt{985}}$ $(= 11.088\ldots)$ **[M1]**

11.1 **[A1]**

[Total Marks: 11]